DUAL PRICES AS LAGRANGE MULTIPLIERS

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Linear Programming Example

- Problem: Product M1 requires 1 kg R_1 and 2 kg R_2. One unit of M_2 requires 3 kg R_1 and 1 kg R_2. Maximum available R_1 is 100 kg and maximum available R_2 is 150 kg. Profit of $M_1$is Rs2/unit and that of M_2 is Rs. 3/unit. Formulate the problem.
- Solution:

Maximize $2 M_1 + 3 M_2$

Subject to

$M_1 + 3 M_2 \leq 100$

$2M_1 + M_2 \leq 150,$

$M_1, M_2 \geq 0$

DUAL

Minimize 100 y_1 + 150 y_2

Subject to

$y_1 + 2\ y_2 \geq 2$

$3y_1 + y_2 \geq 3$,

$y_1,\ y_2 \geq 0$



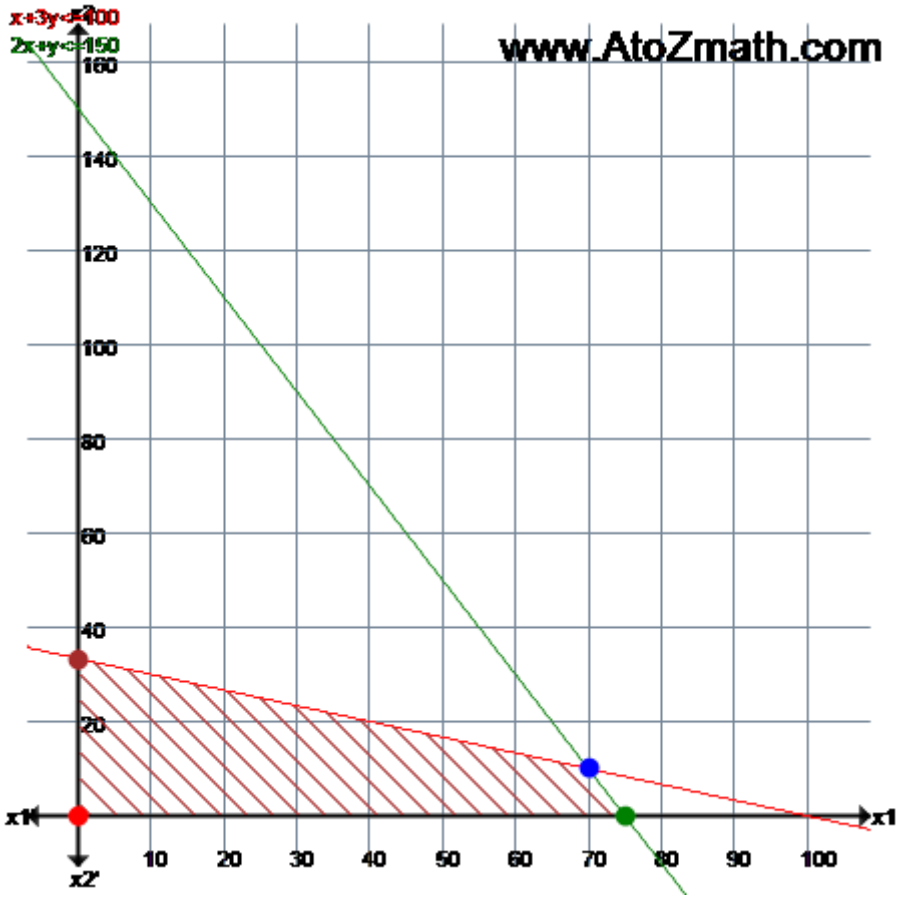
The value of the objective function at each of these extreme points is as follows:

Extreme Point Coordinates (x_1, x_2)	Lines through Extreme Point	Objective function value $Z = 100x_1 + 150x_2$
$A(0, 3)$	$2 \rightarrow 3x_1 + x_2 \geq 3$ $3 \rightarrow x_1 \geq 0$	$100(0) + 150(3) = 450$
$B(0.8, 0.6)$	$1 \rightarrow x_1 + 2x_2 \geq 2$ $2 \rightarrow 3x_1 + x_2 \geq 3$	$100(0.8) + 150(0.6) = 170$
$C(2, 0)$	$1 \rightarrow x_1 + 2x_2 \geq 2$ $4 \rightarrow x_2 \geq 0$	$100(2) + 150(0) = 200$

The miniimum value of the objective function $Z = 170$ occurs at the extreme point $(0.8, 0.6)$.

Hence, the optimal solution to the given LP problem is : $x_1 = 0.8, x_2 = 0.6$ and min $Z = 170$.

Maximize $2\ M_1 + 3\ M_2$
Subject to
$M_1 + 3\ M_2 \leq 100$
$2M_1 + M_2 \leq 150,$
$M_1, M_2 \geq 0$



The value of the objective function at each of these extreme points is as follows:

Extreme Point Coordinates (x_1, x_2)	Lines through Extreme Point	Objective function value $Z = 2x_1 + 3x_2$
$O(0, 0)$	$3 \rightarrow x_1 \geq 0$ $4 \rightarrow x_2 \geq 0$	$2(0) + 3(0) = 0$
$A(75, 0)$	$2 \rightarrow 2x_1 + x_2 \leq 150$ $4 \rightarrow x_2 \geq 0$	$2(75) + 3(0) = 150$
$B(70, 10)$	$1 \rightarrow x_1 + 3x_2 \leq 100$ $2 \rightarrow 2x_1 + x_2 \leq 150$	$2(70) + 3(10) = 170$
$C(0, 33.33)$	$1 \rightarrow x_1 + 3x_2 \leq 100$ $3 \rightarrow x_1 \geq 0$	$2(0) + 3(33.33) = 100$

The maximum value of the objective function $Z = 170$ occurs at the extreme point $(70, 10)$.

Hence, the optimal solution to the given LP problem is : $x_1 = 70, x_2 = 10$ and max $Z = 170$.

Maximize $2M_1 + 3M_2$
Subject to
$M_1 + 3M_2 \leq 100$
$2M_1 + M_2 \leq 150$,
$M_1, M_2 \geq 0$

Constraints $M_1 + 3M_2 \leq 100$ and $2M_1 + M_2 \leq 150$ are active.

$L = 2M_1 + 3M_2 + \lambda_1(M_1 + 3M_2 - 100) + \lambda_2(2M_1 + M_2 - 150)$

$\frac{\partial L}{\partial M_1} = 2 + \lambda_1 + 2\lambda_2 = 0$ — (1)

$\frac{\partial L}{\partial M_2} = 3 + 3\lambda_1 + \lambda_2 = 0$ — (2)

$\frac{\partial L}{\partial \lambda_1} = M_1 + 3M_2 - 100 = 0$ — (3)

$\frac{\partial L}{\partial \lambda_2} = 2M_1 + M_2 - 150 = 0$ — (4)

solving (3) & (4), we get $M_1 = 70$, $M_2 = 30$

solving (1) & 2, we get

$\lambda_1 = -0.8$, $\lambda_2 = -0.6$

They are dual prices with sign changed.

Interpretation of Lagrange Multipliers

Minimize $f(\boldsymbol{x})=0$,

s.t. $\underset{\sim}{h}(\boldsymbol{x}) = b$, where $h(\boldsymbol{x}) = b - \underset{\sim}{h}(\boldsymbol{x})$.

The necessary conditions to be satisfied for the solution of the problem:

$$\frac{\partial f}{\partial x_i} + \lambda \frac{\partial h}{\partial x_i} = 0,\ i = 1,2,\ldots..,n$$

$$h = 0$$

Now,

$$\mathrm{d}b - \mathrm{d}\underset{\sim}{h} = 0, \text{ or } \mathrm{d}b - \sum_{i=1}^{n} \frac{\partial \underset{\sim}{h}}{\partial x_i} \mathrm{d}x_i = 0 \quad \text{or}$$

$$\mathrm{d}b - \sum_{i=1}^{n} \frac{1}{\lambda} \frac{\partial f}{\partial x_i} \mathrm{d}x_i = 0$$

Or

$$\mathrm{d}b = \frac{\mathrm{d}f}{\lambda}$$

Hence,

$$\lambda = \mathrm{d}f / \mathrm{d}b$$

In linear programming, the dual price of a constraint, also known as the shadow price or Lagrange multiplier, represents the sensitivity of the optimal objective function value to changes in the constraint's right-hand side (RHS) value. Essentially, it indicates how much the optimal solution value would change if relaxed or tightened the constraint by a small amount

Lagrange Multipliers:

In mathematical optimization, Lagrange multipliers are used to find the local maxima and minima of a function subject to equality constraints. They effectively introduce a new variable (the multiplier) for each constraint, allowing one to incorporate the constraint into the optimization process

Dual prices are crucial for sensitivity analysis, allowing you to understand how changes in resources or constraints might affect the optimal solution

Minimize $x+y$

s.t.

$$x+2y-6=0$$

$$L=(x+y)+\lambda(x+2y-6)$$

$$\frac{\partial L}{\partial x}=1+\lambda=0 \Rightarrow \lambda=-1$$

$$\frac{\partial L}{\partial y}=1+2\lambda=0 \Rightarrow \lambda=-1/2$$

$$\frac{\partial L}{\partial \lambda}=x+2y-6=0$$

There is no unique value of λ. It means there is no solution to problem. Infact it is unbounded. Minimum can approach negative infinity.

Maximize $3x+4y$

s.t. $x+y=10$

$L = 3x+4y+\lambda(10-x-y)$

$\frac{\partial L}{\partial x} = 3-\lambda = 0 \Rightarrow \lambda = 3$

$\frac{\partial L}{\partial y} = 4-\lambda = 0 \Rightarrow \lambda = 4$

$\frac{\partial L}{\partial \lambda} = 10-x-y = 0$

Here also, we do not get a solution. Infact, solution is unbounded.

How?

Maximize $3x+4y = 3x+4(10-x)$

$= 40-x$

As x is unrestricted, we can take as large a negative value of x as we wish.

Find solution using Simplex method (BigM method)
MIN Z = x1 + x2
subject to
x1 + 2x2 <= 6
and x1,x2 unrestricted in sign

Solution:
Problem is

Min $Z = x_1 + x_2$

subject to

$x_1 + 2x_2 \leq 6$

and x_1, x_2 unrestricted in sign

Since x_1, x_2 are unrestricted in sign, introduce the non-negative variables x_1', x_1'', x_2', x_2''

so that $x_1 = x_1' - x_1'', x_2 = x_2' - x_2''; x_1', x_1'', x_2', x_2'' \geq 0.$

The standard form of the LP problem becomes
Problem is

Min $Z = x_1' - x_1'' + x_2' - x_2''$

subject to

$x_1' - x_1'' + 2x_2' - 2x_2'' \leq 6$

and $x_1', x_1'', x_2', x_2'' \geq 0;$

The problem is converted to canonical form by adding slack, surplus and artificial variables as appropiate

1. As the constraint-1 is of type ' $\leq$ ' we should add slack variable S_1

After introducing slack variables

Min $Z = x_1' - x_1'' + x_2' - x_2'' + 0S_1$

subject to

$x_1' - x_1'' + 2x_2' - 2x_2'' + S_1 = 6$

and $x_1', x_1'', x_2', x_2'', S_1 \geq 0$

Iteration-1		C_j	1	-1	1	-1	0	
B	C_B	X_B	x_1'	x_1''	x_2'	x_2''	S_1	**MinRatio** $\frac{X_B}{x_1''}$
S_1	0	6	1	-1	2	-2	1	---
$Z = 0$		Z_j	0	0	0	0	0	
		$Z_j - C_j$	-1	1 ↑	-1	1	0	

Variable x_1'' should enter into the basis, but all the coefficients in the x_1'' column are negative or zero. So x_1'' can not be entered into the basis.

Hence, the solution to the given problem is unbounded.